Maxwell–Teleparallel Algebra & Chern–Simons Theory

Reference Notes

Maxwell–Teleparallel Gravity

Teleparallel Geometry

Weitzenböck connection $\hat{\Gamma}^{\lambda}{}_{\mu\nu} = e_a^{\lambda}\partial_\mu e_\nu^a$ has *zero curvature, non-zero torsion*:

$$T^a{}_{\mu\nu} = \partial_\mu e_\nu^a - \partial_\nu e_\mu^a.$$

TEGR Lagrangian $T = T^{a\mu\nu}T_{a\mu\nu} - 2T^{a\mu\nu}T_{a\nu\mu} - \frac{1}{2}T^{a\mu}{}_{\mu}T_{a\nu}{}^{\nu}$ reproduces Einstein equations up to boundary term.

Maxwell–Teleparallel Algebra

Combine $[P_a, P_b] = Z_{ab}$ (Maxwell) with vanishing spin connection (teleparallel constraint). Connection and curvatures:

$$A = e^a P_a + \tfrac{1}{2}k^{ab}Z_{ab}, \quad T^a = de^a, \quad F_{(Z)}^{ab} = dk^{ab} + e^a \wedge e^b.$$

CS action in three dimensions:

$$S_{\mathrm{MT-CS}} = \frac{k}{4\pi}\int_{\mathcal{M}_3}\langle A \wedge dA + \tfrac{2}{3}A^3\rangle,$$

where $\langle\cdot,\cdot\rangle$ must pair P_a with Z_{bc} (abelian Z_{ab} forces cross-terms: $\langle P_a, Z_{bc}\rangle \sim \delta_a^{[b}\delta^{c]}$) for non-degeneracy.

Super-CS extension. Promote to superalgebra $\mathfrak{g} = \mathfrak{g}_0 \oplus \mathfrak{g}_1$ with graded bracket $[T_A, T_B\} = f_{AB}{}^{C}T_C$. For $\mathcal{N} = 1$: $\mathfrak{osp}(1|2)$ with $\{Q_\alpha, Q_\beta\} = (\gamma^a C)_{\alpha\beta}J_a$; gravitino ψ^α enters super-connection. Central charge:

$$c = c_{\mathrm{bos}} + c_{\mathrm{ferm}}, \quad c_{\mathrm{bos}} = \frac{k \dim G_{\mathrm{MT}}}{k + h^\vee}.$$

CS/WZW & Holography

CS/WZW: CS on $\mathcal{M}_3$ with $\partial\mathcal{M}_3 = \Sigma$ induces WZW on Σ (impose $A_{\bar{z}}|_\Sigma = 0$). Physical states $\leftrightarrow$ conformal blocks; partition function = topological invariant of $\mathcal{M}_3$.

AdS$_3$/CFT$_2$: 3d gravity with $\Lambda < 0$ is $SL(2,\mathbb{R})^2$ CS theory, $A_\mu^{(\pm)} = \omega_\mu \pm e_\mu/\ell$, level $k = \ell/(4G_N)$. Brown–Henneaux:

$$c = \frac{3\ell}{2G_N} = 6k.$$

Dictionary: bulk flat connections $\leftrightarrow$ boundary gauge fields; Wilson lines in R $\leftrightarrow$ boundary primaries in R; CS on handlebody $\leftrightarrow$ CFT torus partition function.